

Homework 9

68011008 Waroon Ragwongsiri

25/12/2025

1 จงพิสูจน์ว่า $(a \bmod n) = (b \bmod n)$ แล้ว $n \mid (a - b)$

$$\text{let } a = nk_1 + p, p \in \mathbb{Z}_{\geq 0} \quad (1)$$

$$\text{let } b = nk_2 + p, p \in \mathbb{Z}_{\geq 0} \quad (2)$$

$$a - b = (nk_1 + p) - (nk_2 + p) \quad (3)$$

$$a - b = nk_1 + p - nk_2 - p \quad (4)$$

$$a - b = nk_1 - nk_2 \quad (5)$$

$$a - b = n(k_1 - k_2) \quad (6)$$

$$\text{let } x = k_1 - k_2 \quad (7)$$

$$x = \mathbb{Z}_{\geq 0} - \mathbb{Z}_{\geq 0} \quad (8)$$

$$\therefore x \in \mathbb{Z} \quad (9)$$

$$a - b = nx, x \in \mathbb{Z} \quad (10)$$

$$\therefore (a \bmod n) = (b \bmod n) \text{ แล้ว } n \mid (a - b)$$